ClipSAT : Multivariable Calculus Partials, Multiple Integrals & Vector Calcu [ANSWER KEY]

Unit 2.01: Multivariable Chain Rule, Directional Derivatives & Gradients

Q1. Find nabla f-5,3 for fx,y=4 x^{2} + 3 y^{2} .

[ ] A: 18, -40
[ ] B: -20, 9
[x] C: -40, 18
[ ] D: 40, -18

Solution: nabla f=8 x,6 y , so nabla f-5,3=-40, 18 .

Q2. Find nabla f-6,3 for fx,y=3 x^{2} + y^{2} .

[ ] A: 6, -36
[ ] B: -18, 3
[x] C: -36, 6
[ ] D: 36, -6

Solution: nabla f=6 x,2 y , so nabla f-6,3=-36, 6 .

Q3. Find nabla f5,-1 for fx,y=3 x^{2} + 5 y^{2} .

[ ] A: -10, 30
[x] B: 30, -10
[ ] C: 15, -5
[ ] D: -30, 10

Solution: nabla f=6 x,10 y , so nabla f5,-1=30, -10 .

Q4. Find nabla f-5,-5 for fx,y=9 x^{2} + 6 y^{2} .

[x] A: -90, -60
[ ] B: -60, -90
[ ] C: -45, -30
[ ] D: 90, 60

Solution: nabla f=18 x,12 y , so nabla f-5,-5=-90, -60 .

Q5. Find nabla f-2,2 for fx,y=2 x^{2} + 7 y^{2} .

[ ] A: 28, -8
[ ] B: -4, 14
[ ] C: 8, -28
[x] D: -8, 28

Solution: nabla f=4 x,14 y , so nabla f-2,2=-8, 28 .

Q6. Find nabla f-2,-1 for fx,y=4 x^{2} + 8 y^{2} .

[ ] A: -8, -8
[ ] B: 16, 16
[x] C: -16, -16
[ ] D: -16, -14

Solution: nabla f=8 x,16 y , so nabla f-2,-1=-16, -16 .

Q7. Find nabla f-2,-4 for fx,y=2 x^{2} + 6 y^{2} .

[ ] A: -48, -8
[x] B: -8, -48
[ ] C: -4, -24
[ ] D: 8, 48

Solution: nabla f=4 x,12 y , so nabla f-2,-4=-8, -48 .

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Q8. Find nabla f-1,-4 for fx,y=4 x^{2} + 8 y^{2} .

[ ] A: -64, -8

[ ] B: -4, -32

[x] C: -8, -64

[ ] D: 8, 64

Solution: nabla f=8 x,16 y , so nabla f-1,-4=-8, -64 .